

Chapter 3

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1 Convergent Sequences

Def 1. A sequence (p_n) in a metric space X is said to *converge* if there is a point $p \in X$ with the following property: For every $\varepsilon > 0$ there is an integer N such that $n \geq N$ implies that $d(p_n, p) < \varepsilon$.

In this case, we also say that (p_n) converges to p , or that p is the limit of (p_n) , and we write $p_n \rightarrow p$, or

$$\lim_{n \rightarrow \infty} p_n = p.$$

If (p_n) does not converge, it is said to *diverge*.

The set of points p_n ($n = 1, 2, 3, \dots$) is the *range* of (p_n) . The range of a sequence may be a finite set, or it may be infinite. The sequence (p_n) is said to be *bounded* if its range is bounded.

Thm 2. Let (p_n) be a sequence in a metric space X .

(a) (p_n) converges to $p \in X$ if and only if every neighborhood of p contains p_n for all but finitely many n .

(b) If $p \in X$, $p' \in X$, and if (p_n) converges to p and to p' , then $p' = p$.

(c) If (p_n) converges, then (p_n) is bounded.

(d) If $E \subset X$ and if p is a limit point of E , then there is a sequence (p_n) in E such that $p = \lim_{n \rightarrow \infty} p_n$.

Thm 3. Suppose (s_n) , (t_n) are complex sequences and $\lim_{n \rightarrow \infty} s_n = s$, $\lim_{n \rightarrow \infty} t_n = t$. Then

- (a) $\lim_{n \rightarrow \infty} (s_n + t_n) = s + t$;
- (b) $\lim_{n \rightarrow \infty} (cs_n) = cs$, $\lim_{n \rightarrow \infty} (c + s_n) = c + s \quad \forall c$;
- (c) $\lim_{n \rightarrow \infty} (s_n t_n) = st$;
- (d) $\lim_{n \rightarrow \infty} (1/s_n) = 1/s$, provided $s_n \neq 0$ and $s \neq 0$.

Thm 4.

- (a) Suppose $\mathbf{x}_n \in \mathbb{R}^k$ ($n = 1, 2, 3, \dots$) and

$$\mathbf{x}_n = (\alpha_{1,n}, \dots, \alpha_{k,n}).$$

Then $(\mathbf{x}_n)$ converges to $\mathbf{x} = (\alpha_1, \dots, \alpha_k)$ if and only if

$$\lim_{n \rightarrow \infty} \alpha_{j,n} = \alpha_j \quad (1 \leq j \leq k).$$

- (b) Suppose $(\mathbf{x}_n)$, $(\mathbf{y}_n)$ are sequences in $\mathbb{R}^k$, (β_n) is a sequence of real numbers, and $\mathbf{x}_n \rightarrow \mathbf{x}$, $\mathbf{y}_n \rightarrow \mathbf{y}$, $\beta_n \rightarrow \beta$. Then

$$\lim_{n \rightarrow \infty} (\mathbf{x}_n + \mathbf{y}_n) = \mathbf{x} + \mathbf{y}, \quad \lim_{n \rightarrow \infty} \mathbf{x}_n \cdot \mathbf{y}_n = \mathbf{x} \cdot \mathbf{y}, \quad \lim_{n \rightarrow \infty} \beta_n \mathbf{x}_n = \beta \mathbf{x}.$$

2 Subsequences

Def 5. Given a sequence (p_n) , consider a sequence (n_k) of positive integers such that $n_1 < n_2 < n_3 < \dots$. Then the sequence (p_{n_i}) is called a *subsequence* of (p_n) . If (p_{n_i}) converges, its limit is called the *subsequential limit* of (p_n) .

Thm 6.

- (a) If (p_n) is a sequence in a compact metric space X , then some subsequence of (p_n) converges to a point of X .
- (b) Every bounded sequence in $\mathbb{R}^k$ contains a convergent subsequence.

Thm 7. The subsequential limits of a sequence (p_n) in a metric space form a closed subset of X .

3 Cauchy Sequences

Def 8. A sequence (p_n) in a metric space X is said to be a *Cauchy sequence* if for every $\varepsilon > 0$ there is an integer N such that $d(p_n, p_m) < \varepsilon$ if $n > N$ and

$m > N$.

Def 9. Let E be a nonempty subset of a metric space X and let S be the set of all real numbers of the form $d(p, q)$ with $p \in E$ and $q \in E$. The sup of S is called the *diameter* of E .

Thm 10.

(a) If $\overline{E}$ is the closure of a set E in a metric space X , then

$$\text{diam} \overline{E} = \text{diam} E.$$

(b) If K_n is a sequence of compact sets in X such that $K_n \supset K_{n+1}$ ($n = 1, 2, 3, \dots$) and if

$$\lim_{n \rightarrow \infty} \text{diam} K_n = 0,$$

then $\bigcap_{n=1}^{\infty} K_n$ consists of exactly one point.

Thm 11.

(a) In any metric space X , every convergent sequence is a Cauchy sequence.

(b) If X is a compact metric space and if (p_n) is a Cauchy sequence in X , then (p_n) converges to some point of X .

(c) In $\mathbb{R}^k$, every Cauchy sequence converges.

Def 12. A metric space in which every Cauchy sequence converges is said to be *complete*.

Def 13. A sequence (s_n) of real numbers is said to be

(a) *monotonically increasing* if $s_n \leq s_{n+1}$ ($n = 1, 2, 3, \dots$);

(b) *monotonically decreasing* if $s_n \geq s_{n+1}$ ($n = 1, 2, 3, \dots$).

The class of monotonic sequences consists of the increasing and decreasing sequences.

Thm 14. Suppose (s_n) is monotonic. Then (s_n) converges if and only if it is bounded.

4 Upper and Lower Limits

Def 15. Let (s_n) be a sequence of real numbers with the following property: For every real M there is an integer N such that $n \geq N$ implies $s_n \geq M$. We then write

$$s_n \rightarrow +\infty.$$

Similarly, if for every real M there exists an integer N such that $n \geq N$ implies $s_n \leq M$, we write

$$s_n \rightarrow -\infty.$$

Def 16. Let (s_n) be a sequence of real numbers. Let E be the set of numbers x (in the extended real number system) such that $s_{n_k} \rightarrow x$ for some subsequence (s_{n_k}) . This set E contains all subsequential limits.

Put

$$s^* = \sup E, s_* = \inf E.$$

The numbers s^*, s_* are called the *upper* and *lower* limits of (s_n) ; we use the notation

$$\limsup_{n \rightarrow \infty} s_n = s^*, \quad \liminf_{n \rightarrow \infty} s_n = s_*.$$

Thm 17. Let (s_n) be a sequence of real numbers. Let E and s^* have the same meaning as defined above. Then s^* has the following two properties:

(a) $s^* \in E$.

(b) If $x > s^*$, there is an integer N such that $n > N$ implies $s_n < x$.

Moreover, s^* is the only number with the properties (a) and (b).

Thm 18. If $s_n \leq t_n$ for $n \geq N$, where N is fixed, then

$$\begin{aligned} \liminf_{n \rightarrow \infty} s_n &\leq \liminf_{n \rightarrow \infty} t_n, \\ \limsup_{n \rightarrow \infty} s_n &\leq \limsup_{n \rightarrow \infty} t_n. \end{aligned}$$

5 Some Special Sequences

Thm 19.

(a) If $p > 0$, then $\lim_{n \rightarrow \infty} 1/n^p = 0$.

(b) If $p > 0$, then $\lim_{n \rightarrow \infty} \sqrt[p]{p} = 1$.

(c) $\lim_{n \rightarrow \infty} \sqrt[p]{n} = 1$.

(d) If $p > 0$ and α is real, then $\lim_{n \rightarrow \infty} n^\alpha / (1 + p)^n = 0$.

(e) If $|x| < 1$, then $\lim_{n \rightarrow \infty} x^n = 0$.

6 Series

Def 20. Given a sequence (a_n) , we use the notation

$$\sum_{n=p}^q a_n \quad (p \leq q)$$

to denote the sum $a_p + a_{p+1} + \dots + a_q$. With (a_n) , we associate a sequence (s_n) , where

$$s_n = \sum_{k=1}^n a_k.$$

For (s_n) we also use the symbolic expression

$$a_1 + a_2 + a_3 + \dots$$

or, more concisely,

$$\sum_{n=1}^{\infty} a_n.$$

The symbol above we call an *infinite series* or just a *series*. The numbers s_n are called the *partial sums* of the series. If (s_n) converges to s , we say that the series *converges*, and write

$$\sum_{n=1}^{\infty} a_n = s.$$

The number s is called the sum of the series; but it should be clearly understood that s is the limit of a sequence of sums, and is not obtained simply by addition.

Thm 21. $\sum a_n$ converges if and only if for every $\varepsilon > 0$ there is an integer N such that

$$\left| \sum_{k=n}^m a_k \right| \leq \varepsilon$$

if $m \geq n \geq N$.

Thm 22. If $\sum a_n$ converges, then $\lim_{n \rightarrow \infty} a_n = 0$.

Thm 23. A series of nonnegative terms converges if and only if its partial sums form a bounded sequence.

Thm 24.

(a) If $|a_n| \leq c_n$ for $n \geq N_0$, where N_0 is some fixed integer, and if $\sum c_n$ converges, then $\sum a_n$ converges.

(b) If $a_n \geq d_n \geq 0$ for $n \geq N_0$ and if $\sum d_n$ diverges, then $\sum a_n$ diverges.

7 Series of Nonnegative Terms

Thm 25. If $0 \leq x < 1$, then

$$\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}.$$

If $x \geq 1$, the series diverges.

Thm 26. Suppose $a_1 \geq a_2 \geq a_3 \geq \dots \geq 0$. Then the series $\sum_{n=1}^{\infty} a_n$ converges if and only if the series

$$\sum_{k=0}^{\infty} 2^k a_{2^k} = a_1 + 2a_2 + 4a_4 + 8a_8 + \dots$$

converges.

Thm 27. $\sum \frac{1}{n^p}$ converges if $p > 1$ and diverges if $p \leq 1$.

Thm 28. If $p > 1$,

$$\sum_{n=2}^{\infty} \frac{1}{n(\log n)^p}$$

converges; if $p \leq 1$, the series diverges.

8 The Number e

Def 29. $e = \sum_{n=0}^{\infty} \frac{1}{n!}$.

Thm 30. $\lim_{n \rightarrow \infty} (1 + \frac{1}{n})^n = e$.

Thm 31. e is irrational.

9 The Root and Ratio Tests

Thm 32. Given $\sum a_n$, put $\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|a_n|}$.

- (a) If $\alpha < 1$, $\sum a_n$ converges;
- (b) If $\alpha < 1$, $\sum a_n$ diverges;
- (c) If $\alpha = 1$, the test gives no information.

Thm 33. The series $\sum a_n$

- (a) converges if $\limsup_{n \rightarrow \infty} |a_{n+1}/a_n| < 1$,
- (b) diverges if $|a_{n+1}/a_n| \geq 1$ for all $n \geq n_0$, where n_0 is some fixed integer.

Thm 34. For any sequence (c_n) of positive numbers,

$$\liminf_{n \rightarrow \infty} \frac{c_{n+1}}{c_n} \leq \liminf_{n \rightarrow \infty} \sqrt[n]{c_n},$$

$$\limsup_{n \rightarrow \infty} \sqrt[n]{c_n} \leq \limsup_{n \rightarrow \infty} \frac{c_{n+1}}{c_n}.$$

10 Power Series

Def 35. Given a sequence (c_n) of complex numbers, the series

$$\sum_{n=0}^{\infty} c_n z^n$$

is called a *power series*. The numbers c_n are called the *coefficients* of the series; z is a complex number.

Thm 36. Given the power series $\sum c_n z^n$, put

$$\alpha = \limsup_{n \rightarrow \infty} \sqrt[n]{|c_n|}, \quad R = \frac{1}{\alpha}.$$

Then $\sum c_n z^n$ converges if $|z| < R$ and diverges if $|z| > R$.

11 Summation by Parts

Thm 37. Given two sequences (a_n) and (b_n) , put

$$A_n = \sum_{k=0}^n a_k$$

if $n \geq 0$; put $A_{-1} = 0$. Then, if $0 \leq p \leq q$, we have

$$\sum_{n=p}^q a_n b_n = \sum_{n=p}^{q-1} A_n (b_n - b_{n+1}) + A_q b_q - A_{p-1} b_p.$$

Thm 38. Suppose

- (a) the partial sums A_n of $\sum a_n$ form a bounded sequence;

(b) $b_0 \geq b_1 \geq b_2 \geq \dots$;

(c) $\lim_{n \rightarrow \infty} b_n = 0$.

Then $\sum a_n b_n$ converges.

Thm 39. Suppose

(a) $|c_1| \geq |c_2| \geq |c_3| \geq \dots$;

(b) $c_{2m-1} \geq 0, c_{2m} \leq 0$ ($m = 1, 2, 3, \dots$);

(c) $\lim_{n \rightarrow \infty} c_n = 0$.

Then $\sum c_n$ converges.

Thm 40. Suppose the radius of convergence of $\sum c_n z^n$ is 1, and suppose $c_0 \geq c_1 \geq c_2 \geq \dots$, $\lim_{n \rightarrow \infty} c_n = 0$. Then $\sum c_n z^n$ converges at every point on the circle $|z| = 1$, except possibly at $z = 1$.

12 Absolute Convergence

Thm 41. If $\sum a_n$ converges absolutely, then $\sum a_n$ converges.

13 Addition and Multiplication of Series

Thm 42. If $\sum a_n = A$ and $\sum b_n = B$, then $\sum (a_n + b_n) = A + B$ and $\sum (ca_n) = cA$ for any fixed c .

Def 43. Given $\sum a_n$ and $\sum b_n$, we put

$$c_n = \sum_{k=0}^n a_k b_{n-k} \quad (n = 0, 1, 2, \dots)$$

and call $\sum c_n$ the *product* of the two given series.

Thm 44. Suppose

(a) $\sum_{n=0}^{\infty} a_n$ converges absolutely,

(b) $\sum_{n=0}^{\infty} a_n = A$,

(c) $\sum_{n=0}^{\infty} b_n = B$,

(d) $c_n = \sum_{k=0}^n a_k b_{n-k}$ ($n = 0, 1, 2, \dots$).

Then

$$\sum_{n=0}^{\infty} c_n = AB.$$

That is, the product of two convergent series converges, and to the right value, if at least one of the two series converges absolutely.

Thm 45. If the series $\sum a_n$, $\sum b_n$, $\sum c_n$ converges to A , B , C , and $c_n = a_0 b_n + \dots + a_n b_0$, then $C = AB$.

14 Rearrangements

Def 46. Let (k_n) , $n = 1, 2, 3, \dots$ be a sequence in which every positive integer appears once and only once (that is, (k_n) is a 1-1 function from $\mathbb{N}$ onto $\mathbb{N}$). Putting

$$a'_n = a_{k_n} \quad (n = 1, 2, 3, \dots),$$

we say that $\sum a'_n$ is a *rearrangement* of $\sum a_n$.

Thm 47. Let $\sum a_n$ be a series of real numbers which converges, but not absolutely. Suppose

$$-\infty \leq \alpha \leq \beta \leq +\infty.$$

Then there exists a rearrangement $\sum a'_n$ with the partial sums s'_n such that

$$\liminf_{n \rightarrow \infty} s'_n = \alpha, \quad \limsup_{n \rightarrow \infty} s'_n = \beta.$$

Thm 48. If $\sum a_n$ is a series of complex numbers which converges absolutely, then every rearrangement of $\sum a_n$ converges, and they all converge to the same sum.

15 Exercise Notes

Thm 49. If (s_n) is a complex sequence, define its arithmetic means σ_n by

$$\sigma_n = \frac{s_0 + s_1 + \dots + s_n}{n+1} \quad (n = 0, 1, 2, \dots).$$

Assume $M < \infty$, $|na_n| \leq M$ for all n , and $\lim \sigma_n = \sigma$. Then $\lim_{n \rightarrow \infty} s_n = \sigma$.

Thm 50. If (E_n) is a sequence of closed, nonempty, and bounded sets in a *complete* metric space X , if $E_n \supset E_{n+1}$, and if

$$\lim_{n \rightarrow \infty} \text{diam } E_n = 0,$$

then $\bigcap_{n=1}^{\infty} E_n$ consists exactly one point.

Thm 51. Completion of metric space X .

Step1. Take all Cauchy sequences in X and sort them into equivalence classes if they have the same limits.

Step2. Let X^* be the set of all equivalence classes and define

$$\Delta(P, Q) = \lim_{n \rightarrow \infty} d(p_n, q_n)$$

as the distance function in X^* .

Step3. Every $x \in X$ corresponds to $P_x = (x, x, x, \dots)$ in X^* , in which way we put our original X into X^* .

Then X^* is complete.